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[Manuals](#) / [Brands](#) / [Liberty Manuals](#) / [Media Converter](#) / [IPEX5001](#) / [Installation manual](#) / PDF

LIBERTY IPEX5001 INSTALLATION MANUAL

Quick Links

[Ipex5001 Encoder](#)

[Ipex5002 Decoder](#)

Table of Contents

Important Safety Instructions

Table of Contents

IPEX5001 Encoder

IPEX5002 Decoder

IPEX5001 Front Panel

IPEX5001 Rear Panel

IPEX5002 Front Panel

IPEX5002 Rear Panel

Gigabit Switch Features

Gigabit Switch Size(s)

System Bandwidth

Gigabit Switch Expansion

LAN Cabling

IPEXCB Controller

System Control

Heat and Ventilation

Backup Power

HDMI CEC

Source Content Resolution

Point to Point Installation

Reset Point to Point Installation to Factory Default

Basic Matrix Installation

RS232 Connections

Analog Audio Connections

USB Connections

IR Connections

Matrix, Switch, Distribution Amplifier

Video Wall

Firmware Updates

Device Reset

Troubleshooting

No Picture

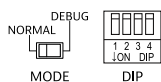
No Audio

Other ManualsLib Projects

IPEX5000 Series Installation Guide

IPEX5001

POWER STATUS



DC 12V



RESET



LAN (PoE)

ANALOG AUDIO



IN



OUT



RS232



USB HOST



HDMI IN



HDMI OUT



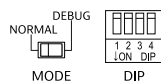
IR IN



IR OUT

IPEX5002

POWER STATUS



USB DEVICE



DC 12V



RESET



LAN (PoE)



AUDIO OUT



RS232



HDMI OUT



IR IN



IR OUT

Important Safety Instructions

1. Read these instructions – All the safety and operating instructions should be read before this product is operated.
2. Keep these instructions – The safety and operating instructions should be retained for future reference.
3. Heed all warnings – All warnings on the appliance and in the operating instructions should be adhered to.
4. Follow all instructions – All operating and use instructions should be followed.
5. Do not use this apparatus near water – The appliance should not be used near water or moisture – for example, in a wet basement or near a swimming pool, and the like.
6. Clean only with a dry cloth.
7. Do not block any ventilation openings. Install in accordance with the manufacturer's instructions.
8. Do not install near any heat sources such as radiators, heat registers, stoves, or other apparatus (including amplifiers) that produce heat.
9. Do not defeat the safety purpose of the polarized plug. A polarized plug has two blades with one wider than the other. The wide blade or the third prong is provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
10. Protect the power cord from being walked on or pinched particularly at the plugs, convenience receptacles, and at the point where it exits from the apparatus.
11. Only use attachments/accessories specified by the manufacturer.
12. Use only with the cart, stand, tripod, bracket, or table specified by the manufacturer, or sold with the apparatus. When a cart or rack is used, use caution when moving the cart/ apparatus combination to avoid injury from tip-over.
13. Unplug the apparatus during lightning storms or when unused for long periods of time.
14. Refer all servicing to qualified service personnel. Servicing is required when the apparatus has been damaged in any way, such as; the power-supply cord or plug is damaged, liquid has been spilled or objects have fallen into the apparatus, the apparatus has been exposed to rain or moisture, does not operate normally, or has been dropped.
15. CAUTION: Servicing instructions are for use by qualified service personnel only. To reduce the risk of electric shock, do not perform any servicing other than that contained in the operating instructions unless you are qualified to do so.
16. Do not install this equipment in a confined or built-in space such as a book case or similar unit. The equipment must remain in well ventilation conditions. Ventilation should not be impeded by covering the ventilation openings with items such as newspaper, table-cloths, curtains etc.
17. WARNING: Only use attachments/accessories (such as the battery etc.) specified or provided by the manufacturer.
18. WARNING: Refer to the information on the underside of the enclosure for electrical and safety information before installing or operating the apparatus.
19. WARNING: To reduce the risk of fire or electric shock do not expose this apparatus to rain or moisture. The apparatus shall not be exposed to dripping or splashing and objects filled with liquids, such as vases, shall not be placed on apparatus.
20. CAUTION: Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type.
21. WARNING: The battery shall not be exposed to excessive heat such as sunshine, fire or the like.
22. WARNING: The all-pole mains switch located on rear panel is used as the disconnect device, the switch shall remain readily operable.
23. WARNING: DO NOT INGEST BATTERY. CHEMICAL BURN HAZARD.
24. Keep new and used batteries away from children. If the battery compartment does not close securely, stop using the product and keep it away from children.
25. If you think batteries might have been swallowed or placed inside any part of the body, seek immediate medical attention.



26. When the apparatus is not in use or during its relocation, take care of the power cord and plugs; e.g. tie up the power cord with cable tie or similar. The tie must be free from sharp edges and the like that might cause abrasion of the power cord. When put into use again ensure the power cord and plugs are not damaged. If any damage is found the power cord and plugs should be replaced by items either specified by the manufacturer or that have same characteristics as the original items.



27. This lightning flash with arrowhead symbol within an equilateral triangle is intended to alert the user to the presence of non-insulated “dangerous voltage” within the product’s enclosure that may be of sufficient magnitude to constitute a risk of electric shock.

28. WARNING: To reduce the risk of electric shock, do not remove cover (or back) as there are no user-serviceable parts inside. Refer servicing to qualified personnel.



29. The exclamation point within an equilateral triangle is intended to alert the user to the presence of important operating and maintenance instructions in the literature accompanying the appliance.



30. Protective earthing terminal. The apparatus should be connected to a mains socket outlet with a protective earthing connection.

31. CAUTION: To prevent electric shock hazard, replace grille. (CSA 60065, clause 5.3A)

Table of Contents

Important Safety Instructions	3
Product Overview.....	6
IPEX5001 Encoder	6
IPEX5002 Decoder.....	6
Package Contents per Device	6
Front and Rear Panels.....	7
IPEX5001 Front Panel.....	7
IPEX5001 Rear Panel.....	7
IPEX5002 Front Panel.....	8
IPEX5002 Rear Panel.....	8
System Considerations.....	9
Gigabit Switch Features	9
Gigabit Switch Size(s).....	9
System Bandwidth	9
Gigabit Switch Expansion.....	9
LAN Cabling.....	10
IPEXCB Controller	10
System Control	10
Heat and Ventilation	10
Backup Power	10
HDMI CEC.....	10
Source Content Resolution	10
Installation Instructions.....	11
Point to Point Installation.....	11
Reset Point to Point Installation to Factory Default	11
Basic Matrix Installation	11
RS232 Connections.....	12
Analog Audio Connections	12
USB Connections	12
IR Connections	12
General Operation.....	14
Matrix, Switch, Distribution Amplifier.....	14
Video Wall.....	15
Updates and Troubleshooting.....	16
Firmware Updates	16
Device Reset.....	16
Troubleshooting	16
No Picture.....	16
No Audio.....	16
IPEX5001 Technical Specifications.....	18
IPEX5002 Technical Specifications.....	19

Product Overview

IPEX5001 Encoder

The IPEX5001 transmits HDMI video and audio over a gigabit IP network using JPEG2000 encoding with an average variable data rate of 250 Mbps with peaks up to 850 Mbps. The IPEX5001 supports video signals up to 4K at 60 Hz 4:2:0 with HDCP 2.2 and 7.1 multichannel audio. Using a 48 port gigabit switch, up to 45 encoders may be used to create an incredible matrix or switching system within a single installation. An analog audio input port embeds the audio with the video content, such as a DVI video source. An analog audio output port de-embeds the audio from the HDMI content, while still passing the audio to the HDMI output.

The IPEX5001 supports PoE and can be powered remotely by compatible power source equipment, such as a PoE Ethernet switch, eliminating the need for a nearby power outlet. The RS232 port relays commands from the IP controller and operates in a pass-through capacity to control a third party device such as an HD video camera. The USB connection passes keyboard and mouse signals from the IPEX5002 decoder to the IPEX5001 encoder.

The IPEX5001 requires a single IPEX5002 for a point to point installation. When multiple IPEX5001 and/or IPEX5002 devices are in a system, an IPEXCB control box will be required.

IPEX5002 Decoder

The IPEX5002 receives HDMI video and audio over a gigabit IP network using JPEG2000 encoding with an average variable data rate of 250 Mbps with peaks up to 850 Mbps. The IPEX5002 supports video signals up to 4K at 30 Hz 4:4:4 with HDCP 2.2 and 7.1 multichannel audio. The IPEX5002 outputs video up to 4K at 30 Hz 4:4:4 and will scale the content based on the native resolution of the connected display. Using a 48 port gigabit switch, up to 45 decoders may be used to create an incredible matrix or distribution system within a single installation. Depending on the needs of the installation, multiple IPEX5002 devices can be configured to make a video wall up to a 16x16 configuration. An analog audio output port de-embeds the audio from the HDMI content, while still passing the audio to the HDMI output.

The IPEX5002 supports PoE and can be powered remotely by compatible power source equipment, such as a PoE Ethernet switch, eliminating the need for a nearby power outlet. The RS232 port relays commands from the IP controller and operates in a pass-through capacity to control a third party device such as a projector. The USB connection passes keyboard and mouse signals from the IPEX5002 decoder to the IPEX5001 encoder.

The IPEX5001 requires a single IPEX5002 for a point to point installation. When multiple IPEX5001 and/or IPEX5002 devices are in a system, an IPEXCB control box will be required.

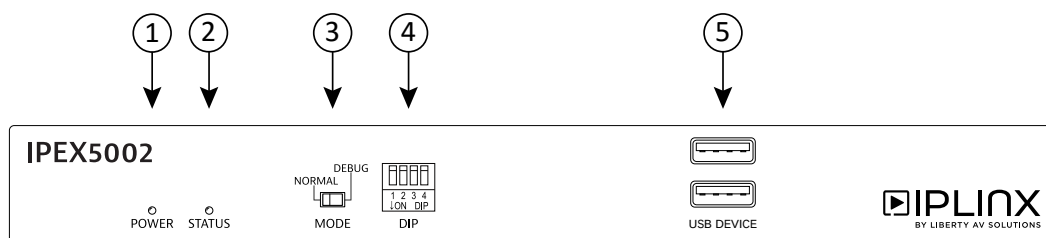
Package Contents per Device

1. Installation Guide
2. Power Supply with US, UK, EU, and AU adapters
3. 3-pin Removable Screw Terminal
4. Mounting Ears (2 ea)
5. IR Emitter
6. IR Receiver

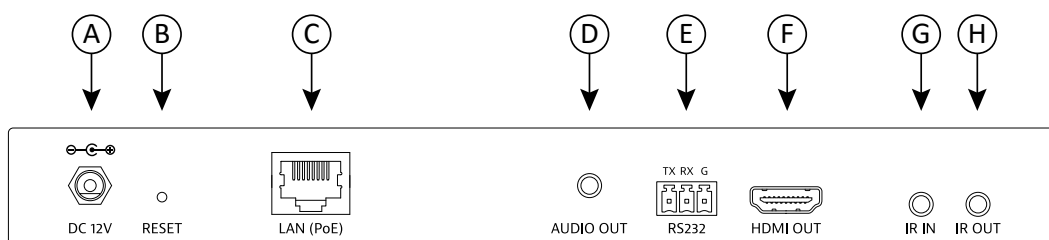
Front and Rear Panels

IPEX5001 Front Panel



IPEX5002 Front Panel

1. Power indicator
2. Status indicator
3. RS232 Function Switch
4. DIP Switch for ID Modes
5. USB ports for keyboard or mouse

IPEX5002 Rear Panel

- A. 12V DC power input
- B. RESET button
- C. LAN connection with PoE support
- D. Analog audio output
- E. RS232 connection
- F. HDMI output
- G. IR input
- H. IR output

System Considerations

Gigabit Switch Features

A high quality, managed Level 2 gigabit switch with a high bandwidth backplane should be used in the installation, preferably with PoE support. The two primary features to look for in the switch are support for multicast with jumbo frames and support for IGMP snooping, which are required technologies for stable video signals.

Be sure to verify the PoE gigabit switch can provide 15.4 watts to each output under full load. Some switches support 15.4 watts per output but do not have enough available power under full load. For a 24 port switch, it should be able to provide about 370 watts to the LAN ports ($24 \times 15.4 = 369.6$) plus power for the switch.

Gigabit Switch Size(s)

When calculating the size of switch needed, the following devices need to be considered:

1. Number of sources
2. Number of displays
3. IPEXCB
4. Wireless access point for iPad app or Windows app (optional)
5. Hardwire port for Windows app (optional)

If the installation requires 14 sources and 10 displays, a 24 port switch will not have enough ports for the installation, because there are no open ports for the IPEXCB.

System Bandwidth

An average of 250 Mbps should be considered when calculating the bandwidth for a 4k/30 video stream. An average of 150 Mbps should be considered when calculating the bandwidth for a 1080p/60 video stream. These values cover the encoded stream plus packet overhead. A source with a 4k/30 video stream and a second source with a 1080p/60 video stream will use 400 Mbps of bandwidth on the backplane of the switch.

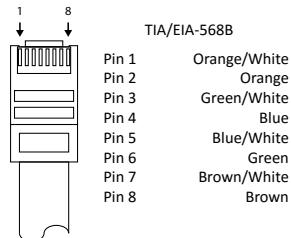
Due to the nature of JPEG2000 compression, the IPEX5001 encoding method is VBR. VBR is Variable Bit Rate. In a VBR stream, the encoder will transmit different bit rates of data depending on how much information is present in the particular frame of video, while averaging the stream to the bandwidth hardcoded into the chipset. A solid color frame will use less bandwidth than a multicolor frame of video.

Gigabit Switch Expansion

The IPEX5001 and IPEX5002 do not currently support expanding the port capacity beyond a single gigabit switch. This is actively being worked on and should be enabled in a firmware update.

LAN Cabling

For all LAN cabling, the EIA/TIA-568B crimp pattern must be used on Category 5e or greater cable. In areas with large amounts of electromagnetic (EM) or radio frequency (RF) interference, a shielded variety of Category 5e or greater cable is recommended with shielded connectors on both ends of the selected cable.



IPEXCB Controller

In addition to the gigabit switch, the IPEXCB is required for the encoders and decoders to communicate with each other regardless of system layout, such as one to many.

System Control

There are multiple ways to control the system once installation is complete. There is the IPLink Control software for iPad and Windows. While not recommended, the IPEXCB can be controlled through a web browser. A third party controller can communicate directly with the IPEXCB via the LAN (CONTROL) port or via RS232.

Heat and Ventilation

All electronics generate heat, and excessive heat can cause electronic devices to fail prematurely. The IPEX5001 and IPEX5002 devices are passively cooled and should not be stacked on top of each other or other devices. Please leave at least 1 ½ inches (approximately 35-40 mm) of open space by the side vents to provide adequate air flow.

The gigabit switch will likely have active cooling. Please follow the manufacturer's installation instructions for proper mounting in an equipment rack.

Backup Power

It is always a good practice to install a high quality UPS (uninterruptible power supply) with line filtering with expensive electronics. The UPS can provide "clean" power to all the devices in the equipment rack while also providing enough time to properly shut down connected devices in the event of an extended power failure.

HDMI CEC

The IPEX5002 can turn on and turn off connected displays via CEC. This technology has been a part of the HDMI specification for years. The IPLinx Control software for Windows and iPad can turn on and off the displays for easy end user control.

Source Content Resolution

Even though the IPEX5000 series devices support 1080i, this resolution may cause artifacts through the encoding process. Please set the output resolution of devices showing 1080i content to 720p or 1080p.

Installation Instructions

Point to Point Installation

1. Connect Category 5E or greater twisted pair cable with the TIA/EIA-568B crimp pattern between the LAN port on the encoder (IPEX5001) and the decoder (IPEX5002).
2. Set the DIP switches on the front of the IPEX5001 and IPEX5002 to the same value, other than all up.
3. Connect the included power supply to the 12V DC power input of the encoder and decoder.
4. Connect an HDMI cable between the display and the decoder (IPEX5002).
5. Connect an HDMI cable between the source and the encoder (IPEX5001).
6. Power on attached audio/video devices.
7. Connect the included power supply connected to the encoders and decoders to an AC outlet.

This method will also work with a one to many installation.

Reset Point to Point Installation to Factory Default

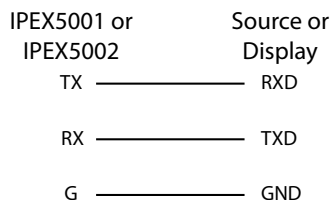
1. Disconnect power from the IPEX5001 and IPEX5002.
2. Set the DIP switches on the front of the IPEX5001 and IPEX5002 to all up.
3. Continue with Basic Matrix Installation instructions.

Basic Matrix Installation

1. Configure the gigabit switch for IP video operation. Liberty has guides for many common switch manufacturers on the IPEXCB page on the Liberty website (www.libav.com).
2. Turn off power and disconnect the audio/video equipment by following the manufacturer's instructions.
3. Turn off power to the configured switch.
4. Connect Category 5E or greater twisted pair cable with the TIA/EIA-568B crimp pattern between the LAN port on the encoder(s) (IPEX5001) and the gigabit switch.
5. Connect Category 5E or greater twisted pair cable with the TIA/EIA-568B crimp pattern between the LAN port on the decoder(s) (IPEX5002) and the gigabit switch.
6. If the gigabit switch cannot provide power or enough power to the IPEX5001 or IPEX5002, connect the included power supply to the 12V DC power input of the device. If the gigabit switch cannot provide enough power, disable the PoE function of the connected LAN ports on the switch.
7. Connect an HDMI cable between the display and the decoder (IPEX5002).
8. Connect an HDMI cable between the source and the encoder (IPEX5001).
9. Power on attached audio/video devices.
10. Apply power to the gigabit switch.
11. If the gigabit switch cannot provide power to the IPEX5001 or IPEX5002, connect the included power supply connected to the encoders and decoders to an AC outlet.

RS232 Connections

To use the RS232 control transport capabilities of the IPEXCB or the RS232 extension capabilities of the IPEX5001 or IPEX5002, connect the TX, RX, and ground control signal wires to the removable 3-pole terminal block. Consult the manual of the source or display device(s) to determine which pins the TX and RX signals are carried on. Be sure to always connect TX to RX and RX to TX.



Analog Audio Connections

The analog audio input and output connections on the IPEX5001 and the analog audio output connections on the IPEX5002 use standard 3.5 mm TRS connectors.

USB Connections

The USB Host port on the IPEX5001 is provided to allow control of a PC source from a remote location.

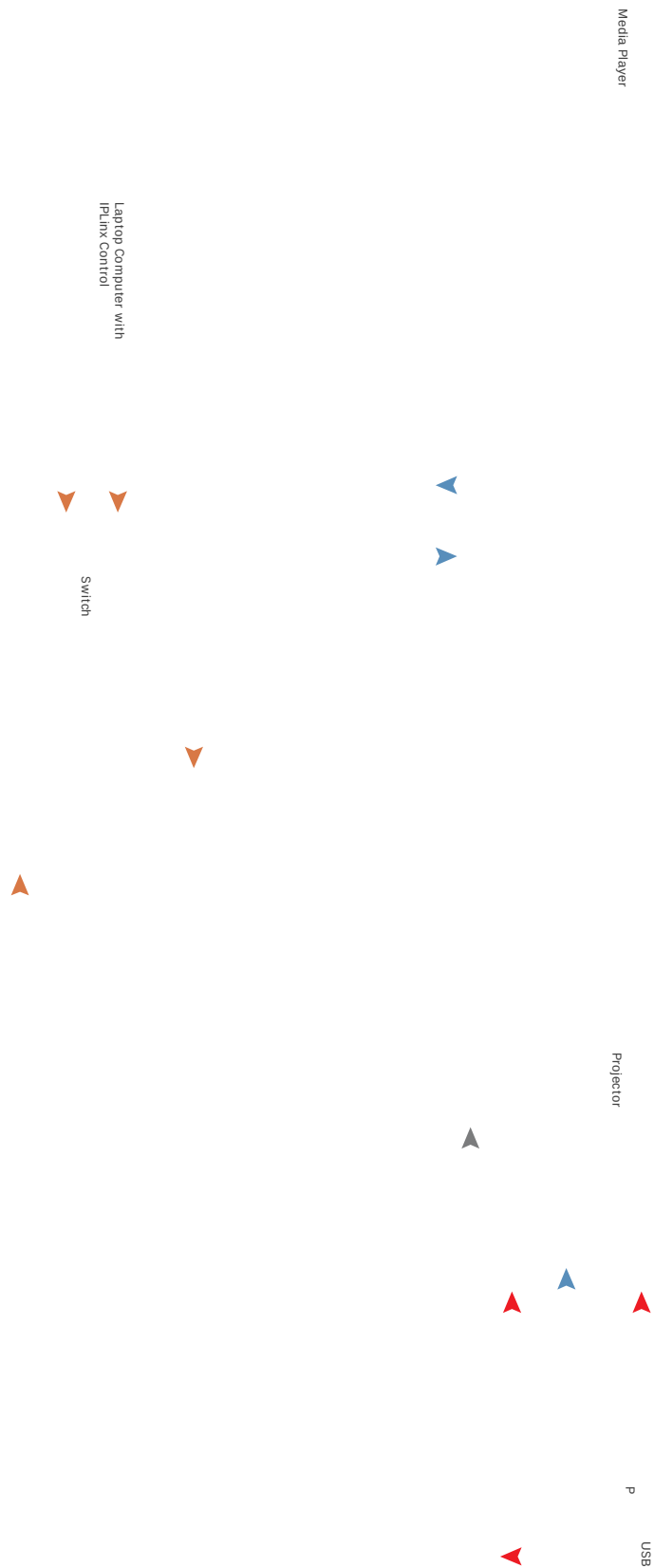
The USB Device ports on the IPEX5002 are provided to add keyboard and mouse control for a host PC.

The USB signal from the IPEX5002 to the IPEX5001 follows the video route.

IR Connections

The IR connections on the IPEX5001 and IPEX5002 provide a means to control remote IR equipment. The IR signals between the IPEX5001 and IPEX5002 follow the video route.

Please ensure power is disconnected from the encoders and decoders before connecting the IR receiver to the IR input ports on the devices.

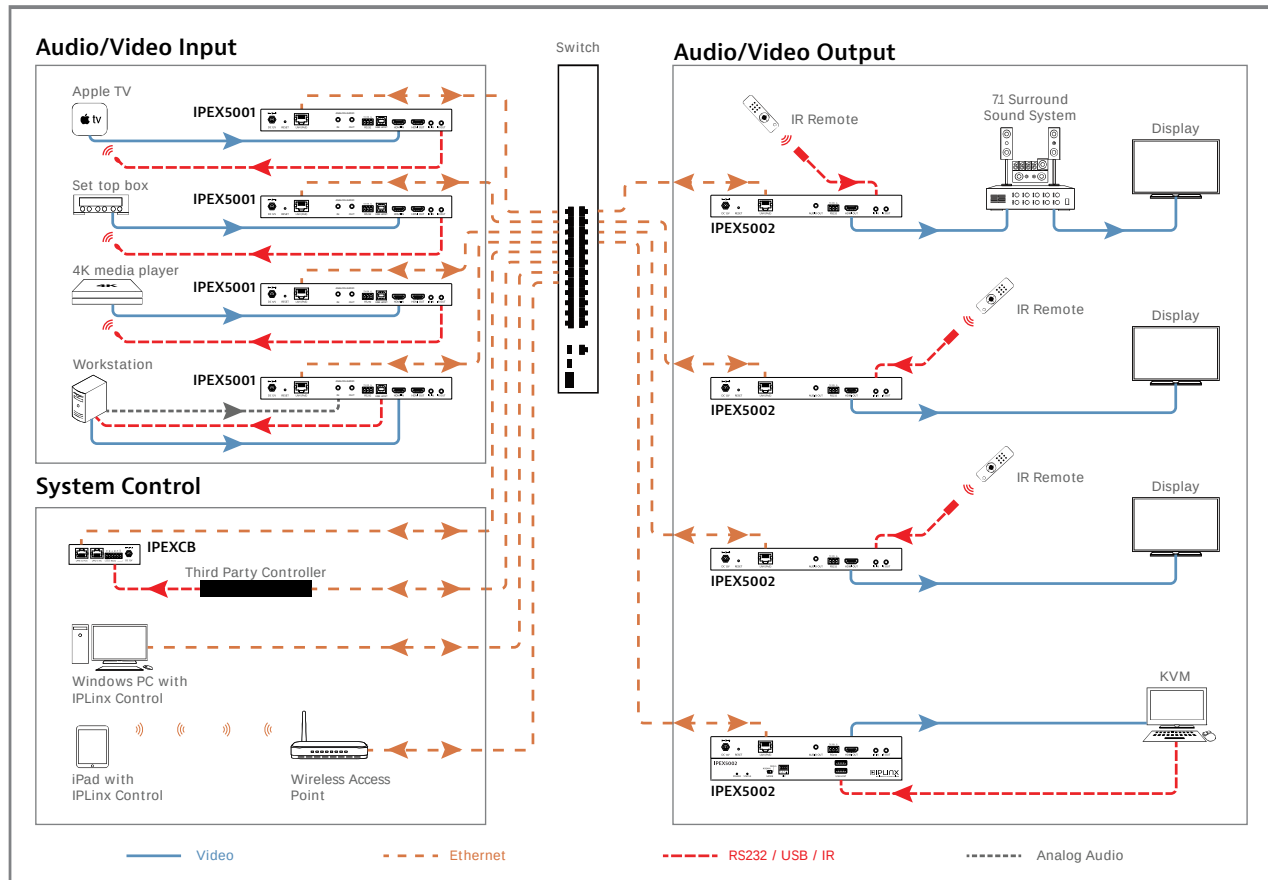


General Operation

Matrix, Switch, Distribution Amplifier

With multiple encoders and decoders, an IPEX5000 series installation can be configured to be a video matrix, video switch, and a distribution amplifier. The video configurations and signal routes can be easily changed through the IPLinx Control software for Windows and iPad or through a third party controller.

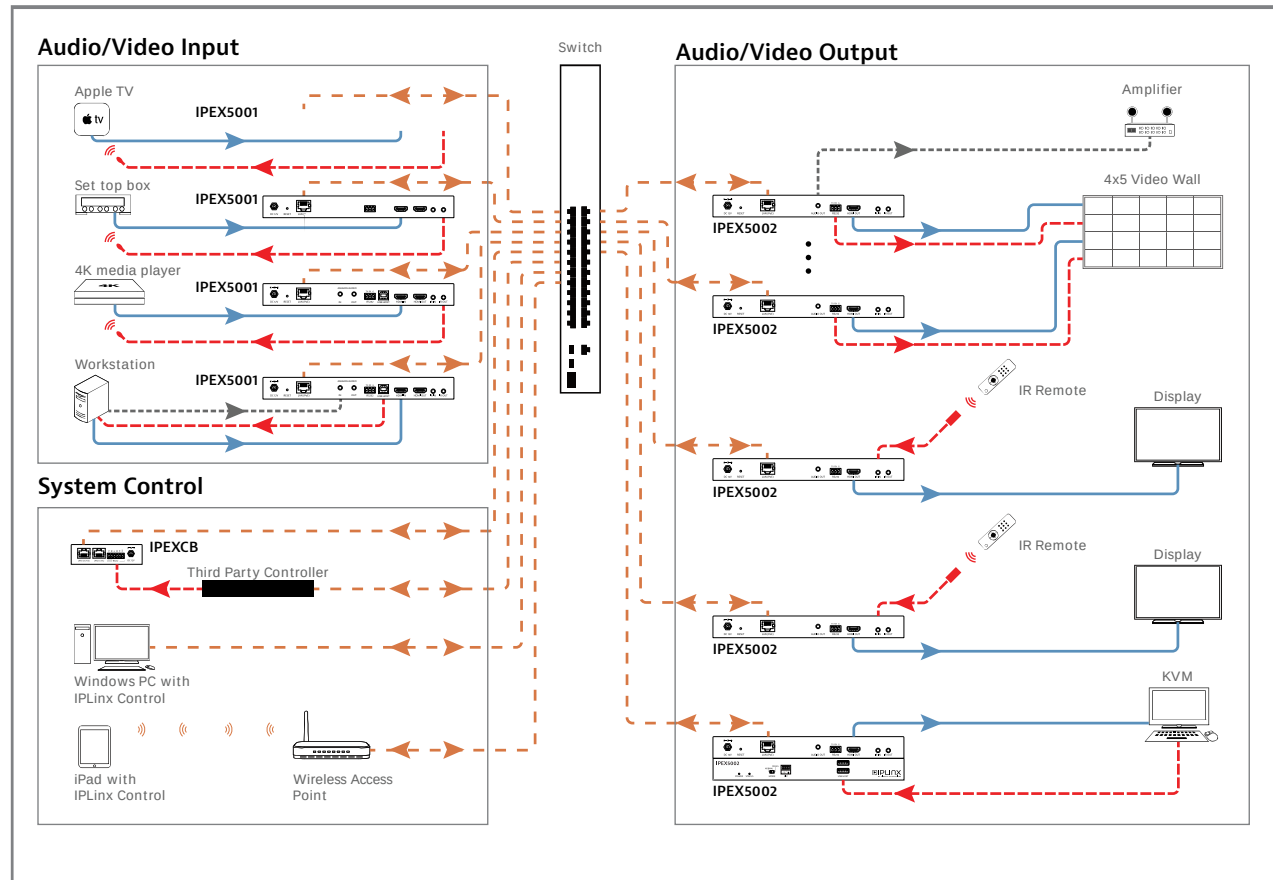
Please see the IPEXCB user manual for configuring this functionality.



Video Wall

In addition to matrix configurations, multiple IPEX5002 devices can be configured to make a video wall up to 16x16 in size. This functionality is also managed through the IPLinx Control software for Windows and iPad or through a third party controller.

Please see the IPEXCB user manual for configuring this functionality.



Updates and Troubleshooting

Firmware Updates

Firmware updates for the IPEX5001 and IPEX5002 are handled through the IPLinx Updater software, which can be found on the IPEXCB page of the Liberty website. Please see the documentation included with the software to update the encoders and decoders.

Device Reset

Resetting the IPEX5001 and IPEX5002 is handled through the IPLinx Config software. Please see the documentation included with the software to reset the encoders and decoders to factory defaults.

Troubleshooting

No Picture

1. Verify all devices are powered on.
2. Verify LAN cable continuity between devices.
3. Verify the STATUS LED is lit on the encoders and decoders.
4. Verify the encoders and decoders are configured properly in the IPLinx Config software.
5. Verify the source and display devices work correctly without signal extension.
6. Verify the display is set to the correct input.
7. If the source content is HDCP encrypted, verify the display supports HDCP.
8. Verify the output resolution of the decoder matches the display resolution in the IPLinx Config software.
9. Verify that only one power source is used with the encoders and decoders. using the included power supplies with a PoE switch that is providing PoE will cause stability issues.

No Audio

1. Verify all devices are powered on.
2. Verify LAN cable continuity between devices.
3. Verify the STATUS LED is lit on the encoders and decoders.
4. Verify the encoders and decoders are configured properly in the IPLinx Config software.
5. Verify the source and display devices work correctly without signal extension.
6. Verify the source audio output is set to Stereo output.
7. If the source content is HDCP encrypted, verify the display supports HDCP.
8. Verify the output resolution of the decoder matches the display resolution in the IPLinx Config software.
9. Verify that only one power source is used with the encoders and decoders. using the included power supplies with a PoE switch that is providing PoE will cause stability issues.

IPEX5001 Technical Specifications

Input/Output Connections	
HDMI Input	One (1) HDMI Type A Receptacle
HDMI Output	One (1) HDMI Type A Receptacle
LAN	One (1) 8P8C port (Shielded RJ45)
Power	One (1) 5.5 mm OD, 2.6 mm ID Threaded Barrel
RS232 Port	One (1) 3-pin Removable Terminal Block Connector
USB Device	One (1) USB Type B Port
Audio Input	One (1) 3.5 mm TRS Receptacle
Audio Output	One (1) 3.5 mm TRS Receptacle
IR Input	One (1) 3.5 mm TRS Receptacle
IR Output	One (1) 3.5 mm TRS Receptacle
Reset	One (1) Momentary Push Button
Mode	One (1) Two Position Slider Switch
DIP	One (1) Four Position DIP Switch
Supported Audio, Video and Control	
Video Resolutions	SMPTE: 480p, 576p, 720p, 1080i, 1080p, 2160p/30 (4:4:4), 2160p/60 (4:2:0) VESA: Resolutions up to 1920x1200 Color Depth: 24, 30, 36 bit
Video Compliance	HDMI 1.4 and HDCP 1.4/2.2
Embedded Audio	Up to PCM 8 channel, Dolby Digital True HD, and DTS-HD Master Audio
ARC (Audio Return Channel)	No
HEC (HDMI Ethernet Channel)	No
CEC (Consumer Electronics Control)	Yes
Supported Baud Rates	2400, 4800, 9600, 19200, 38400, 57600, 115200
USB Compliance	HID only
Streaming Signal Characteristics	
Maximum Distance (point to point)	100 m (328 ft)
Cable Requirements	Category 5e or greater with TIA/EIA-568B crimp pattern
Encoding Data Rate	2160p: 250 Mbps average 1080p: 150 Mbps average
Encoding Method	VBR
End to End Latency	17-33 ms (1-2 fps)
Maximum Video Wall Size	16x16
Chassis and Environmental	
Construction	Black Steel
Dimensions (H x W x D)	25 mm x 220 mm x 130.2 mm (0.98in x 8.66 in x 5.13 in)
Operating Temperature	0° to +40° C (+32° to +104° F)
Operating Humidity	20% to 90%, Non-condensing
Storage Temperature	-10° to +60° C (+14° to +140° F)
Storage Humidity	20% to 90%, Non-condensing
Power and Regulatory	
Power Input	12V DC 1A or 48V DC PoE (Power over Ethernet)
Power over Ethernet (PoE) Compatibility	802.3af Alternative A
Power Consumption	6 watts
ESD Protection	8kV air, 4kV contact
Regulatory	FCC, CE, RoHS
Other	
Warranty	2 years
Diagnostic Indicators	Power and Status
Included Accessories	Installation Guide, Power Supply with US, UK, EU and AU adapters, 3-pin Removable Screw Terminal, Mounting Ears (2 ea), IR emitter, IR receiver
IP Controller	IPEXCB
Compatible Decoder	IPEX5002

Distances and picture quality may be affected by cable grade, cable quality, source and destination equipment, RF and electrical interference, and cable patches.

IPEX5002 Technical Specifications

Input/Output Connections	
HDMI Input	One (1) HDMI Type A Receptacle
HDMI Output	One (1) HDMI Type A Receptacle
LAN	One (1) 8P8C port (Shielded RJ45)
Power	One (1) 5.5 mm OD, 2.6 mm ID Threaded Barrel
RS232 Port	One (1) 3-pin Removable Terminal Block Connector
USB Device	Two (2) USB Type A Port
Audio Output	One (1) 3.5 mm TRS Receptacle
IR Input	One (1) 3.5 mm TRS Receptacle
IR Output	One (1) 3.5 mm TRS Receptacle
Reset	One (1) Momentary Push Button
Mode	One (1) Two Position Slider Switch
DIP	One (1) Four Position DIP Switch
Supported Audio, Video and Control	
Video Resolutions	SMPTE: 480p, 576p, 720p, 1080i, 1080p, 2160p/30 (4:4:4), 2160p/60 (4:2:0) VESA: Resolutions up to 1920x1200 Color Depth: 24, 30, 36 bit
Video Compliance	HDMI 1.4/2.0 and HDCP 1.4/2.2
Embedded Audio	Up to PCM 8 channel, Dolby Digital True HD, and DTS-HD Master Audio
ARC (Audio Return Channel)	No
HEC (HDMI Ethernet Channel)	No
CEC (Consumer Electronics Control)	Yes
Supported Baud Rates	2400, 4800, 9600, 19200, 38400, 57600, 115200
USB Compliance	HID only
Streaming Signal Characteristics	
Maximum Distance (point to point)	100 m (328 ft)
Cable Requirements	Category 5e or greater with TIA/EIA-568B crimp pattern
Encoded Data Rate	2160p: 250 Mbps average 1080p: 150 Mbps average
Encoded Method	VBR
End to End Latency	17-33 ms (1-2 fps)
Maximum Video Wall Size	16x16
Chassis and Environmental	
Construction	Black Steel
Dimensions (H x W x D)	25 mm x 220 mm x 130.2 mm (0.98in x 8.66 in x 5.13 in)
Operating Temperature	0° to +40° C (+32° to +104° F)
Operating Humidity	20% to 90%, Non-condensing
Storage Temperature	-10° to +60° C (+14° to +140° F)
Storage Humidity	20% to 90%, Non-condensing
Power and Regulatory	
Power Input	12V DC 1A or 48V DC PoE (Power over Ethernet)
Power over Ethernet (PoE) Compatibility	802.3af Alternative B
Power Consumption	6 watts
ESD Protection	8kV air, 4kV contact
Regulatory	FCC, CE, RoHS
Other	
Warranty	2 years
Diagnostic Indicators	Power and Status
Included Accessories	Installation Guide, Power Supply, 3-pin Removable Screw Terminal, Mounting Ears (2 ea), IR emitter, IR receiver
IP Controller	IPEXCB
Compatible Encoder	IPEX5001

Distances and picture quality may be affected by cable grade, cable quality, source and destination equipment, RF and electrical interference, and cable patches.

IPLinx is a brand of:



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Other ManualsLib Projects



www.manualslib.com



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